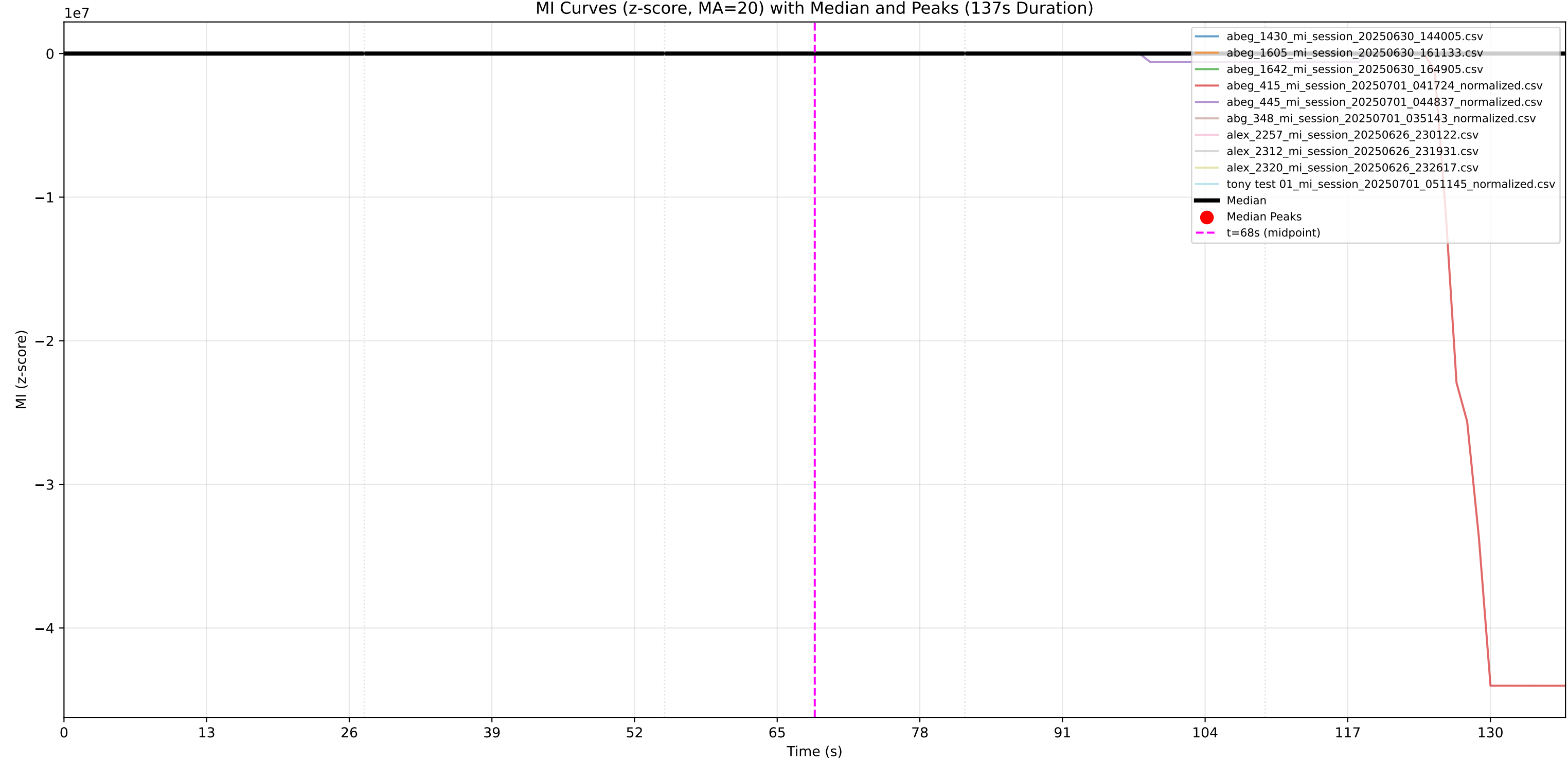
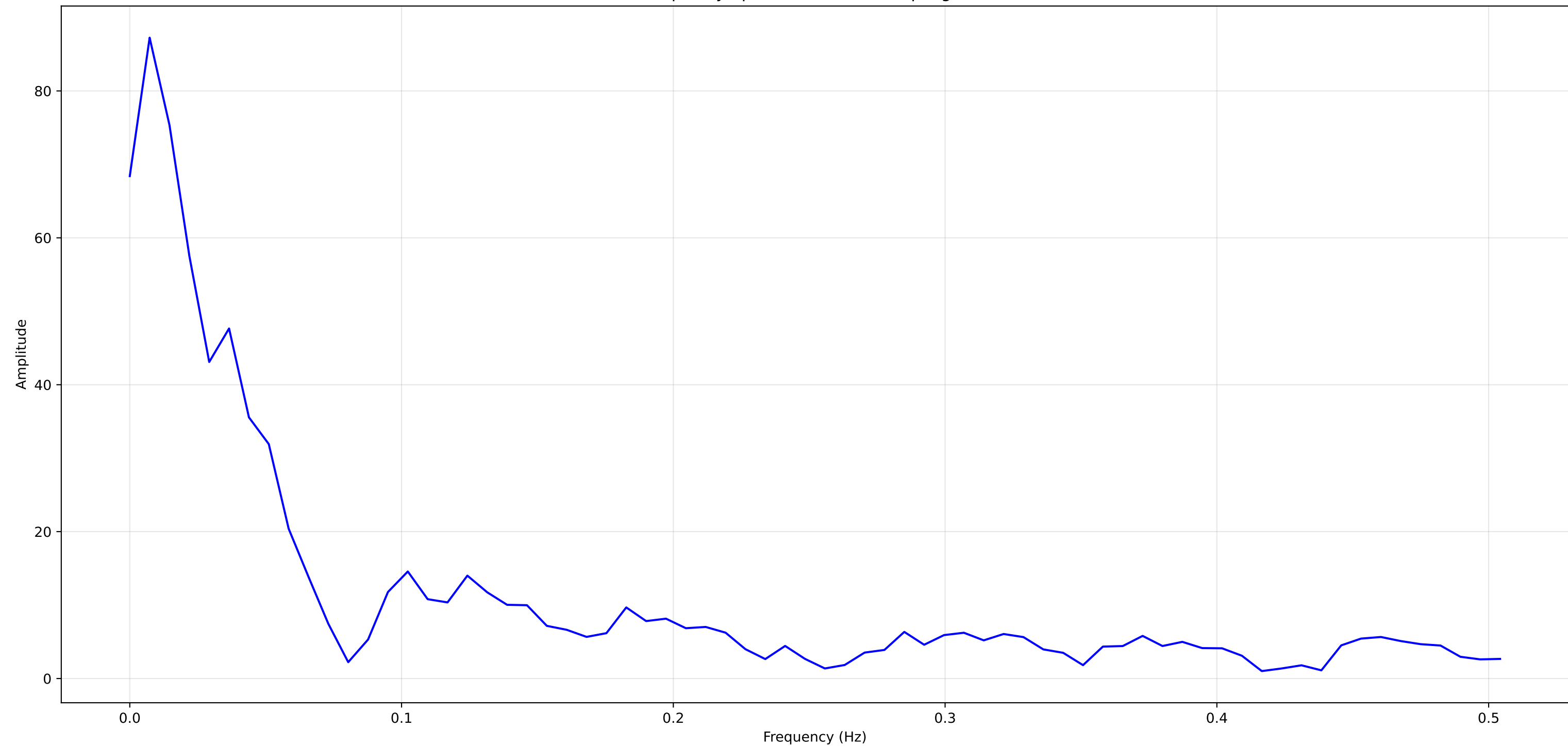


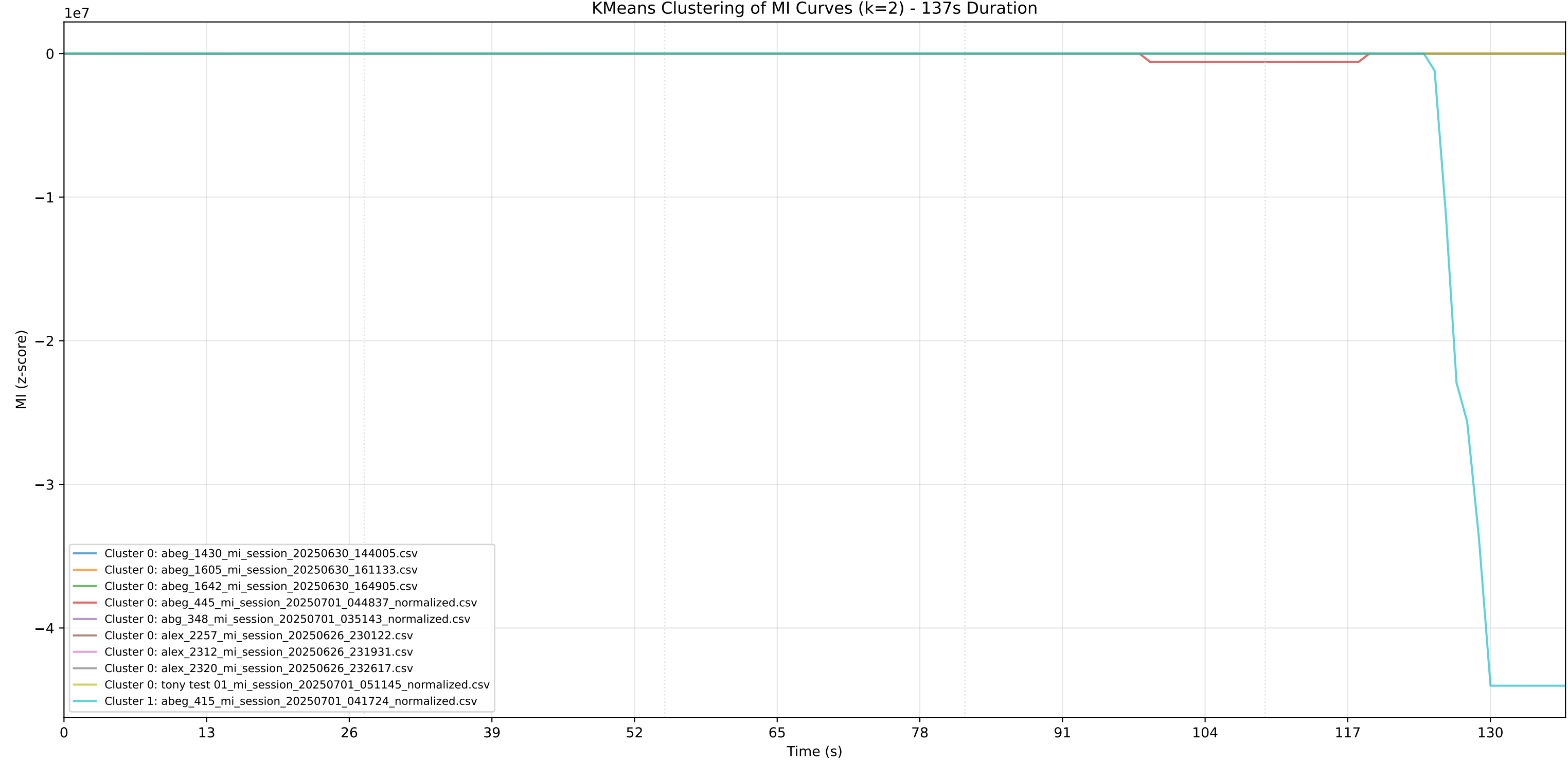
MI Curves (z-score, MA=20) with Median and Peaks (137s Duration)



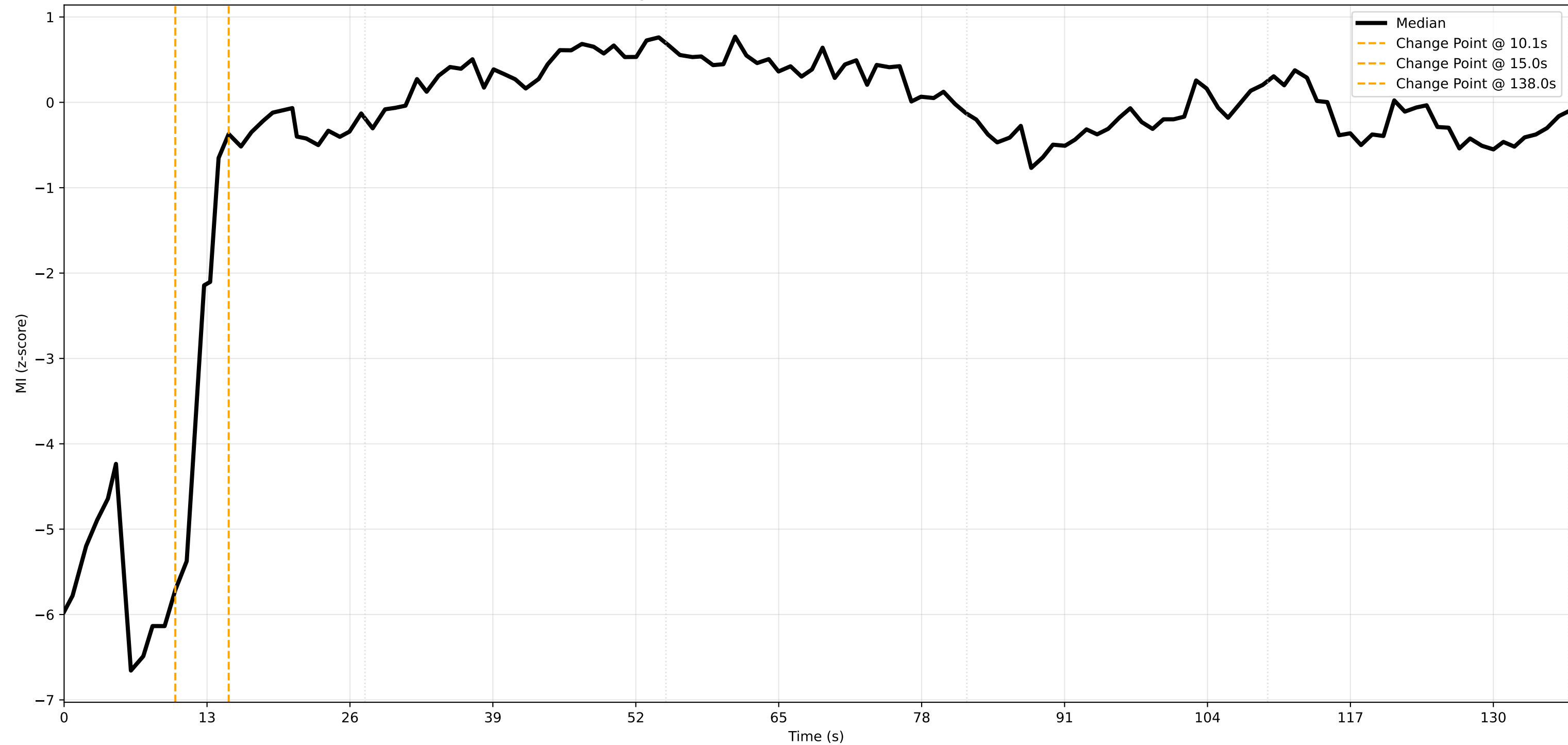
Median MI Frequency Spectrum (FFT) - Sampling Rate: 1.008 Hz



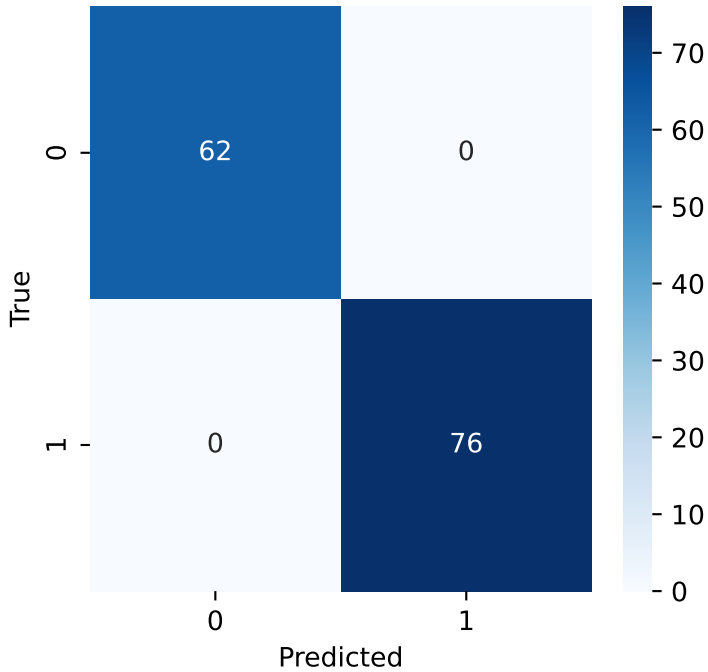
KMeans Clustering of MI Curves (k=2) - 137s Duration



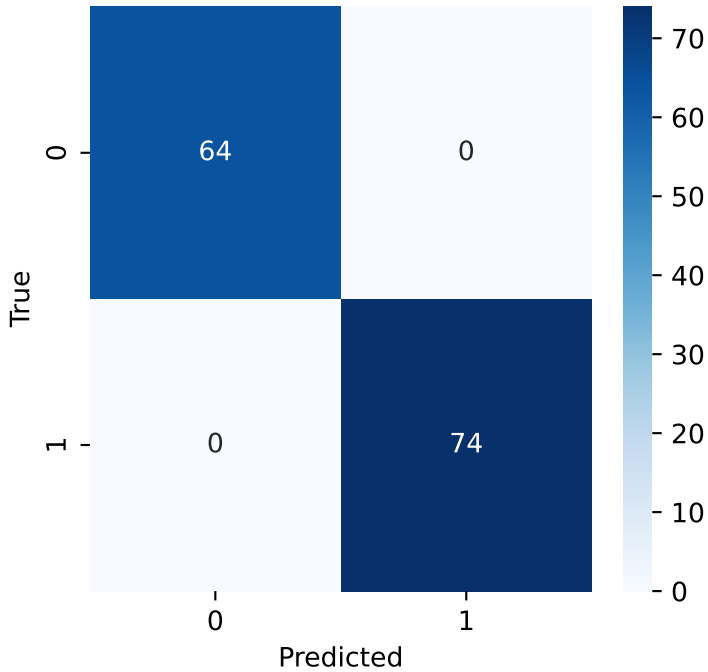
Change Point Detection on Median MI - 137s Duration



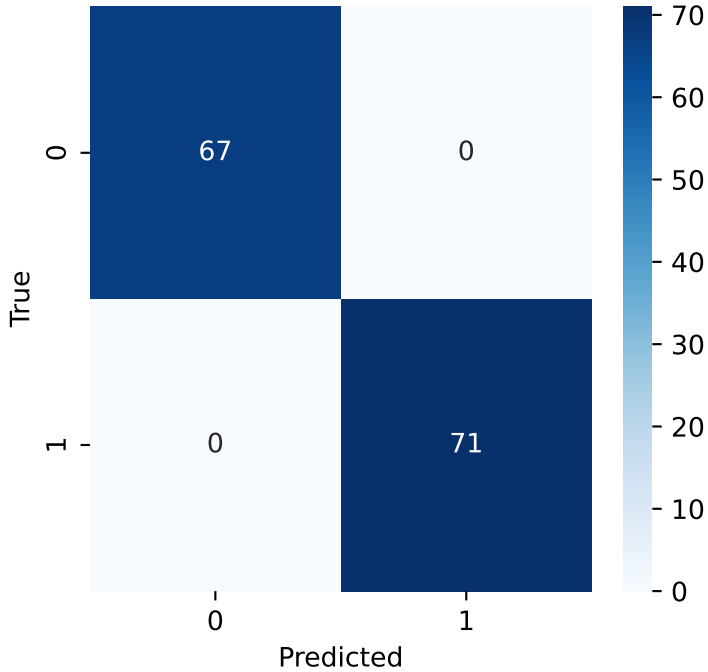
Matrix: abeg_1430_mi_session_20250630_1440



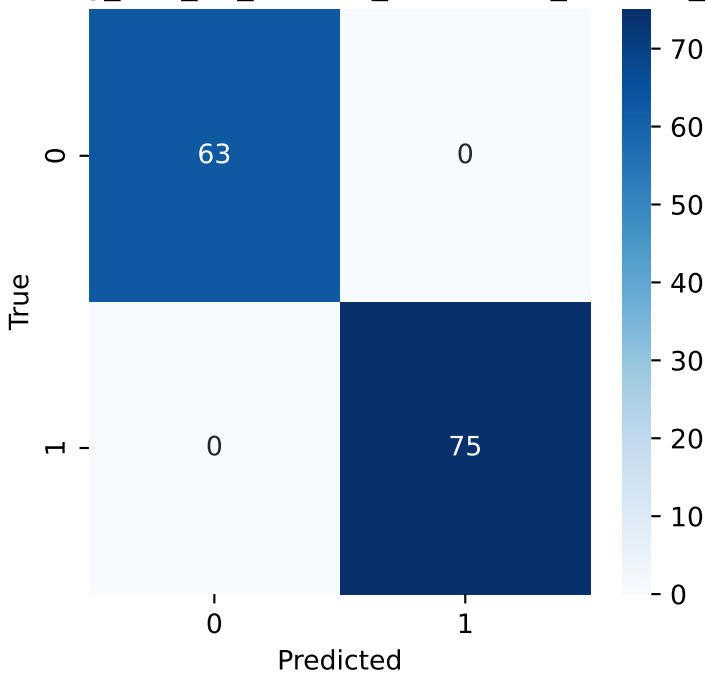
Matrix: abeg_1605_mi_session_20250630_1611



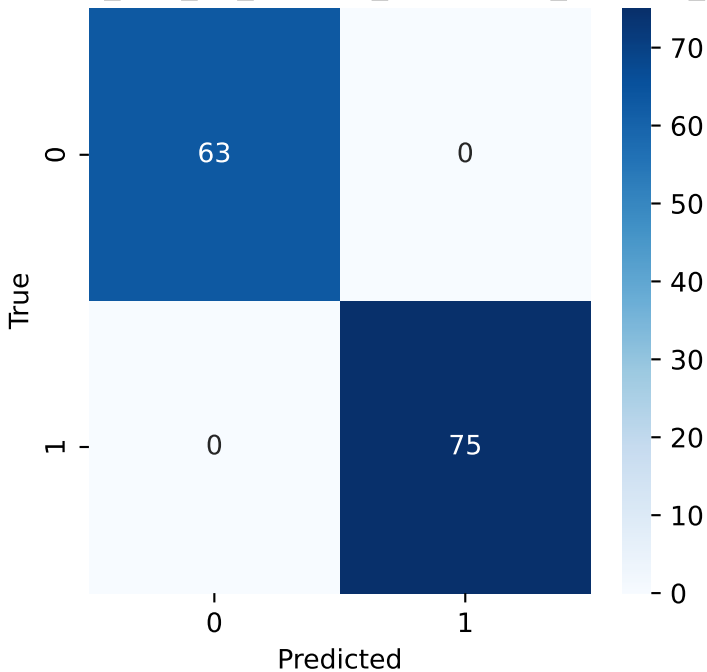
Matrix: abeg_1642_mi_session_20250630_1649



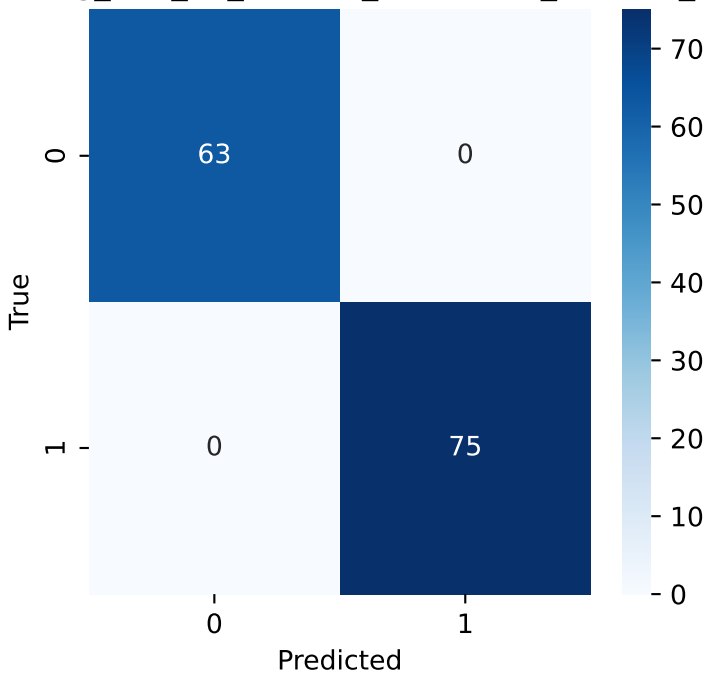
ix: abeg_415_mi_session_20250701_041724_no



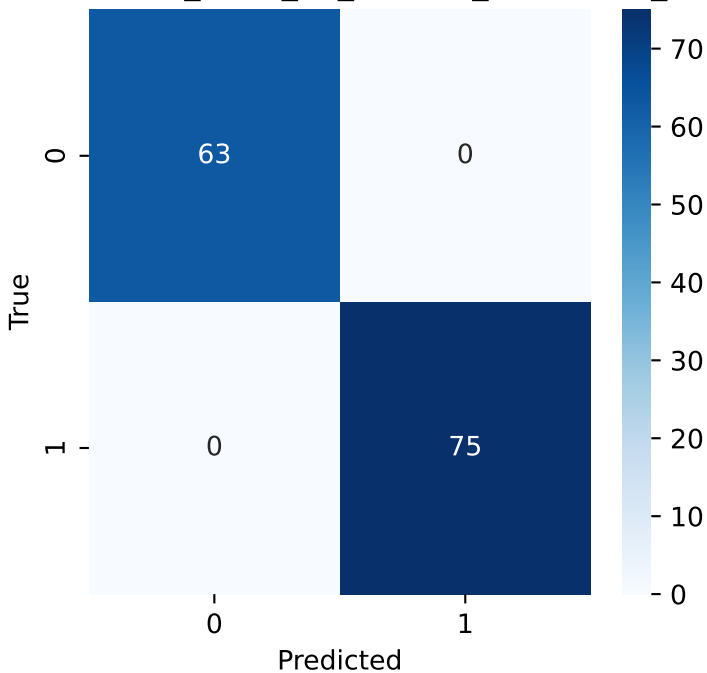
ix: abeg_445_mi_session_20250701_044837_no



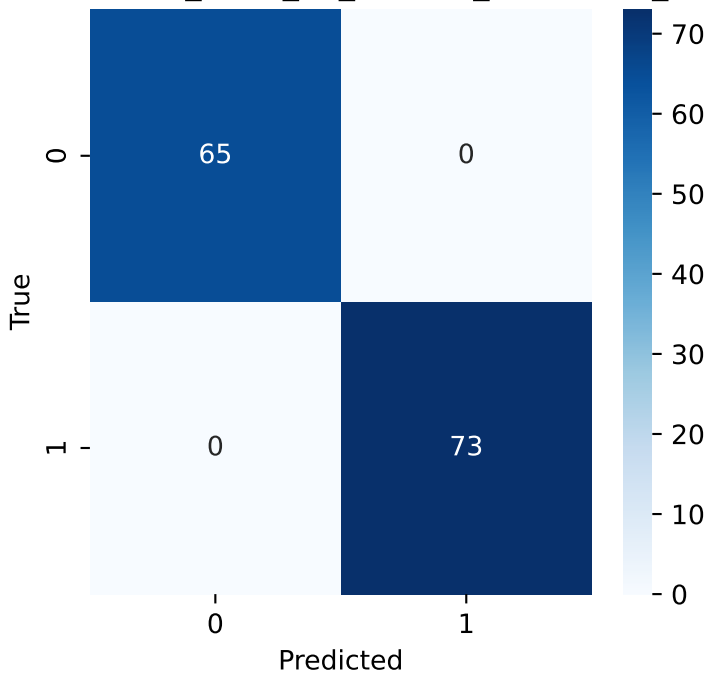
rix: abg_348_mi_session_20250701_035143_nor



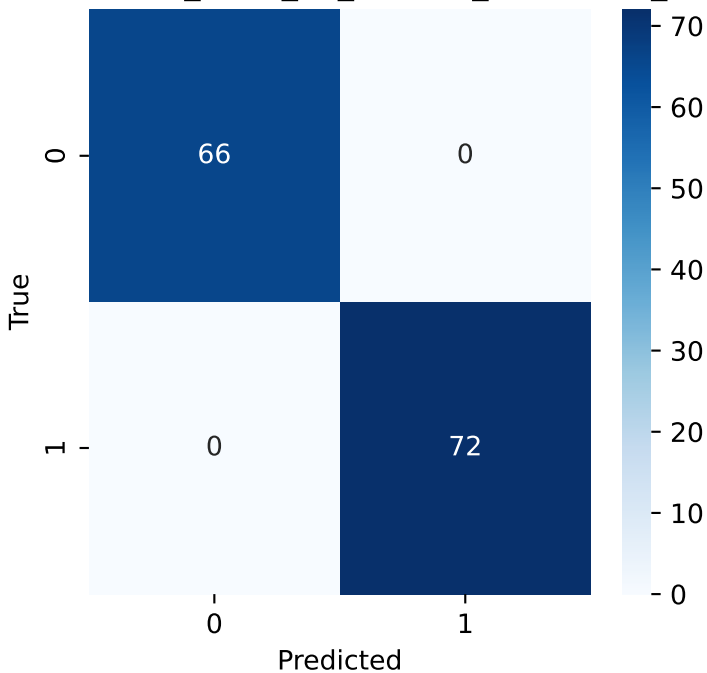
h Matrix: alex_2257_mi_session_20250626_2301



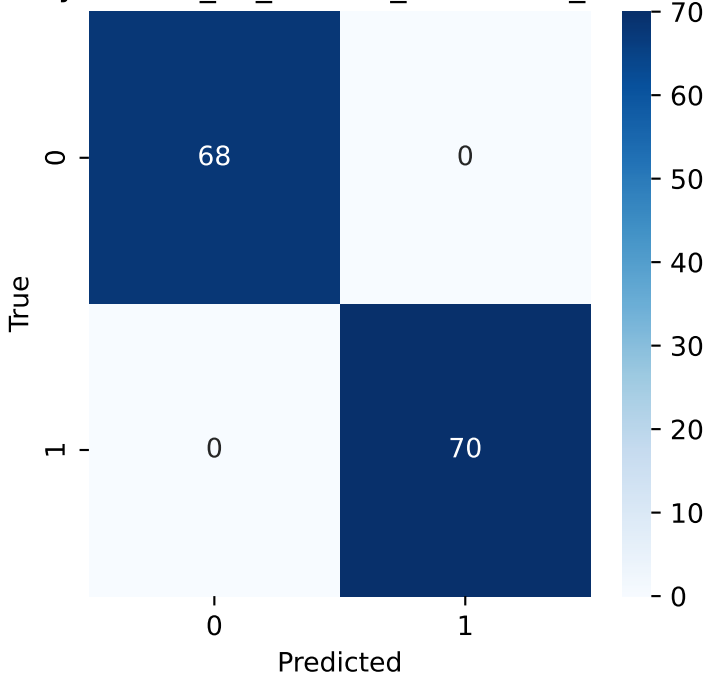
h Matrix: alex_2312_mi_session_20250626_2319



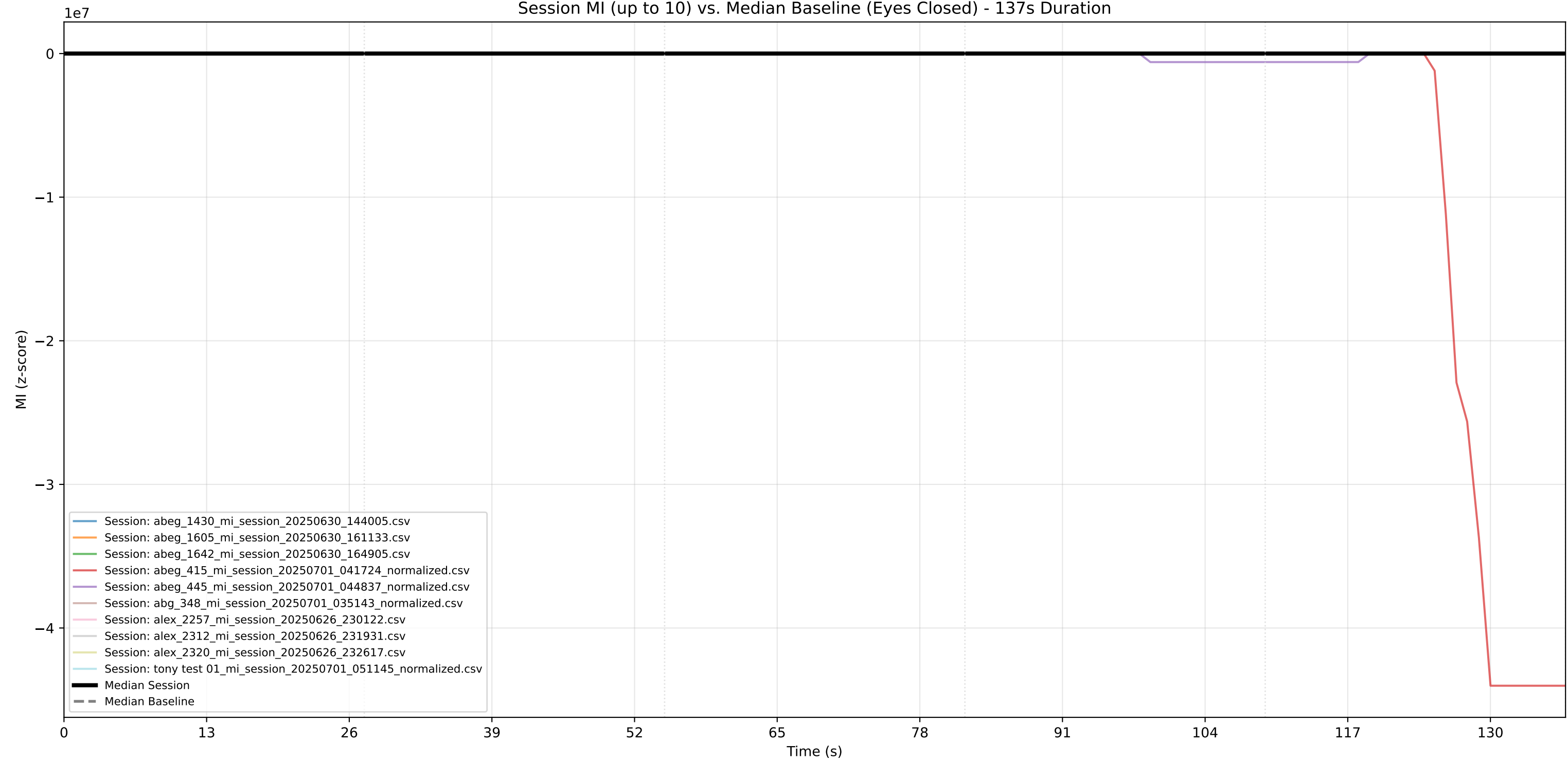
h Matrix: alex_2320_mi_session_20250626_2326



k: tony test 01_mi_session_20250701_051145_n



Session MI (up to 10) vs. Median Baseline (Eyes Closed) - 137s Duration



MI Advanced Report

Generated: 20250717_120315

MI Advanced Report - Statistical Summary (Actual Sampling Rate: 1.008 Hz)

1. Mindfulness Index (MI) Trajectory

- * All MI session files with ≥ 100 samples were included (100s minimum at 1Hz).
- * Individual and median MI curves were plotted with 20-second moving average (see Figure 8.1).
- * Actual data duration: 136.8 seconds with 138 samples
- * Calculated sampling rate: 1.008 Hz
- * RM-ANOVA across six epochs (0-23s, 23-46s, 46-68s, 68-91s, 91-114s, 114-137s):

Anova

```
=====
      F Value Num DF  Den DF Pr > F
-----
Epoch  1.0048 5.0000 45.0000 0.4259
=====
```

* Post-hoc pairwise comparisons (Tukey HSD):

Multiple Comparison of Means - Tukey HSD, FWER=0.05

```
=====
group1  group2  meandiff  p-adj    lower    upper    reject
-----
  0-23s 114-137s -1873206.7452  0.51 -5047385.616 1300972.1257 False
  0-23s  23-46s   722.2935  1.0 -3173456.5773 3174901.1643 False
  0-23s  46-68s   903.9114  1.0 -3173274.9594 3175082.7823 False
  0-23s  68-91s   903.3142  1.0 -3173275.5567 3175082.185  False
  0-23s  91-114s -39751.533  1.0 -3213930.4039 3134427.3378 False
114-137s  23-46s  1873929.0387 0.5095 -1300249.8322 5048107.9095 False
114-137s  46-68s  1874110.6566 0.5094 -1300068.2142 5048289.5274 False
114-137s  68-91s  1874110.0594 0.5094 -1300068.8115 5048288.9302 False
114-137s  91-114s 1833455.2121 0.5336 -1340723.6587 5007634.083  False
  23-46s  46-68s   181.6179  1.0 -3173997.2529 3174360.4888 False
  23-46s  68-91s   181.0207  1.0 -3173997.8502 3174359.8915 False
  23-46s  91-114s -40473.8265  1.0 -3214652.6974 3133705.0443 False
  46-68s  68-91s    -0.5972  1.0 -3174179.4681 3174178.2736 False
  46-68s  91-114s -40655.4445  1.0 -3214834.3153 3133523.4264 False
  68-91s  91-114s -40654.8472  1.0 -3214833.7181 3133524.0236 False
=====
```

* Time-frequency analysis (FFT) of the median MI curve was performed (see Figure 8.3).

2. MI Dynamics

- * Peak analysis (median curve): 0 peaks ($z > 1.5$) at sample times [], amplitudes [], durations [] (see Table 8.2, Figure 8.4).
- * AUC (median): -65.36; mean $\pm$ SD (individual): -43696923.01 $\pm$ 127085999.57 (see Table 8.3).
- * Bootstrap 95% CI for mean MI difference (0-135s vs 136-270s): mean diff = 0.000, CI = [0.000, 0.000].
- * Change point detection (Pelt algorithm): 2 change points at sample indices [10, 15] (see Figure 8.5).

3. Clustering & Mixed-Effects

* K-Means clustering (k=2): cluster averages {}, user distribution {0: 9, 1: 1} (see Figure 8.6).

* Mixed-effects model (MI ~ time + (1|Subject)): Mixed Linear Model
Regression Results

```
=====
Model:          MixedLM      Dependent Variable:  value
No. Observations: 1380      Method:          REML
No. Groups:       10        Scale:          11778219044971.9297
Min. group size:  138      Log-Likelihood: -22714.4978
Max. group size:  138      Converged:       Yes
Mean group size:  138.0

-----
              Coef.      Std.Err.    z    P>|z|    [0.025    0.975]
-----
Intercept    573585.488 359935.372  1.594  0.111 -131874.878 1279045.855
time         -13228.898  2319.120 -5.704  0.000 -17774.289 -8683.506
=====
```